

Predicting Systemic Delivery Risk in Supply Chains: A Graph Attention Network Approach



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Introduction

In the global landscape, late delivery is a systemic risk that triggers a “bullwhip effect” of stockouts and lost revenue. The current industry standards rely on “spreadsheet-style” math, treating a disruption as an independent event and ignoring how a single congested hub impacts the entire network. Furthermore, the existing “smart” models act as black boxes, prioritizing predictive power over the causal insights needed to understand late deliveries.

Objective

The objective of this project is to transform supply chain modeling from a list of independent rows into a connected web of relational intelligence. By applying a Graph Attention Network (GAT) to a real-world e-commerce supply chain dataset, the model uses self-attention to distinguish localized flukes from systemic defects. This approach identifies directional links to root causes, such as congested ports or shared carriers, thereby shifting logistics from reactive firefighting to proactive, visually guided risk mitigation to reduce operational costs and customer churn.

Methodology

- The research utilizes the **Olist Brazilian E-commerce Dataset**, acquired via the Kaggle open-access repository. This dataset comprises real-world commercial records of approximately 100K orders placed across multiple Brazilian marketplaces between 2016 and 2018; it captures the full order lifecycle including price, freight, status, and customer location.
- The orders are modeled as graph nodes, with edges established through a topological sparsification strategy to optimize both computational efficiency and signal relevance. An edge E_{ij} is instantiated if 2 orders share a common destination and node j is among the $k=5$ nearest neighbors of node i based on customer shipping coordinates. This constraint focuses the attention mechanism on spatially proximal logistics clusters, filtering out noise from geographically distant or unrelated operations.
- Our model utilizes a GATv2 architecture with 8 dynamic attention heads to capture complex, context-dependent logistics relationships. To counter oversmoothing and signal dilution, we employ Residual Feature Injection, merging graph-derived embeddings with raw node features. This hybrid architecture acts as a safeguard, ensuring that the model preserves high-impact, instance-level predictors while leveraging the broader 'relational intelligence' gained from the logistics graph.

Experiments and Results

To validate the efficacy of our graph-based approach, we benchmarked our GAT architecture against a suite of standard tabular ensemble methods and alternate graph configurations. The experiments were conducted using an 80/20 train-test split, with performance measured across Accuracy, F1 Score, and AUC-ROC.

Performance Summary

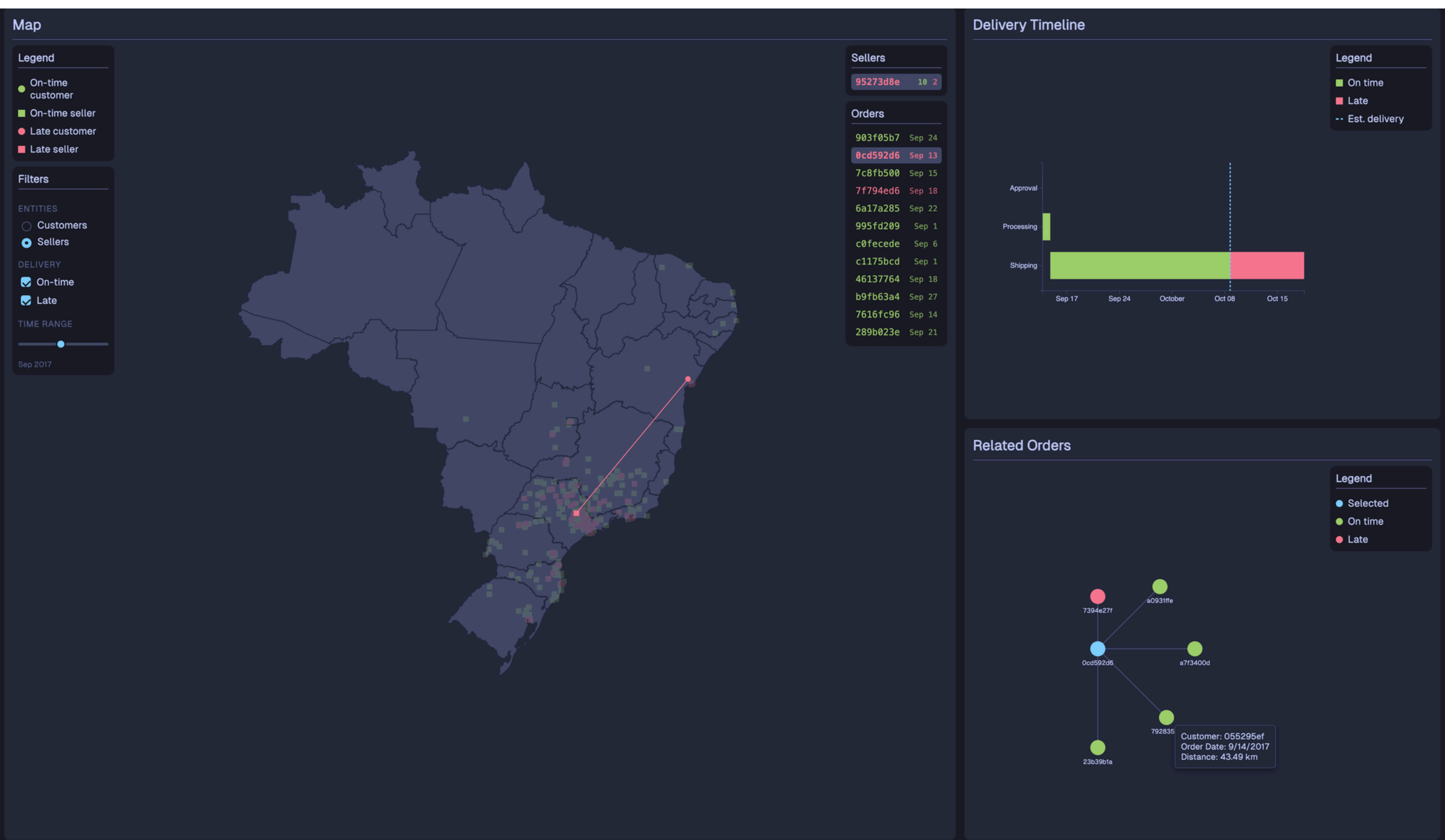
Model	Accuracy	F1 Score (Weighted)	AUC-ROC
Random Forest	94%	92%	74%
XGBoost	78%	83%	71%
GCN	86%	88%	69%
GAT	74%	80%	69%
GNN	87%	88%	70%
MLP	86%	88%	69%

Comparative Analysis

- The **Random Forest** baseline achieves the highest metrics (92% F1, 74% AUC-ROC), confirming that tabular features like seller ID and customer city are highly predictive when treated as independent partitions.
- Our **GNN** and **GCN** models demonstrate strong performance (88% F1), effectively outperforming the XGBoost baseline. This validates our hypothesis: by constructing graph edges based on shared seller ID and customer city attributes, we successfully introduce structural context that traditional models lack.
- The **GAT** (80% F1) trails the top ensemble methods in raw classification performance. However, this is an intentional diagnostic trade-off. Unlike the black-box Random Forest, the GAT’s attention mechanism allows us to map the influence of specific nodes, effectively highlighting why a seller-city pairing is at risk.

Visualization

- We built an **interactive drill-down dashboard** using D3 and React, with 3 goal-oriented and intuitive charts that specifically target our stakeholders.
- The dashboard contains an **order map**, **delivery timeline** and a **related orders** graph.
- The drill-down approach uses a **three-level hierarchy**: locations, customers/sellers, and orders to resolve selection difficulties caused by overlapping markers on the map.
- The **order map** visualizes the **relational innovations** of our model that differentiate it from traditional ML, such as order location and time relationships. It uses position encoding for location and a time-window filter to group concurrent orders.
- The **delivery timeline** shows the entire delivery process for an order to easily surface any bottlenecks in the process.
- The **related orders** chart is an **innovation** that solves the black box model interpretability issue by illustrating the exact k-nearest neighbors formula that our model uses to make predictions. Edge length encodes destination distance between related orders and a tooltip displays order date to indicate temporal relationship.



Conclusion

This project reframes supply chain modeling as a connected graph rather than isolated transactions by applying a GAT to e-commerce data, enabling the detection of systemic issues. While traditional tabular models like Random Forest achieved higher predictive performance, graph-based models demonstrated the added value of structural context, capturing how disruptions propagate across the network. The GAT’s lower accuracy reflects a deliberate trade-off for interpretability, allowing identification of root causes rather than just predictions. It is highly likely that simple geographic proximity is insufficient for modeling complex dependencies, suggesting future improvements through richer edge definitions (e.g., temporal flows, shared hubs), multiclass classification, and inclusion of external factors, such as seasonality and weather, to better capture supply chain dynamics.